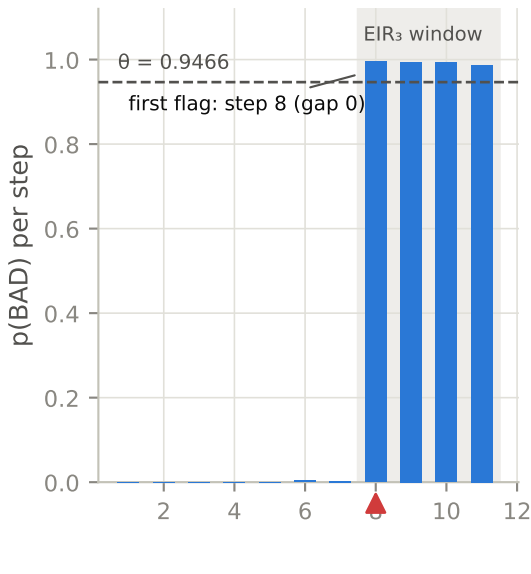


Step-level scoring on held-out benchmark trajectories (frozen model, deployed θ)

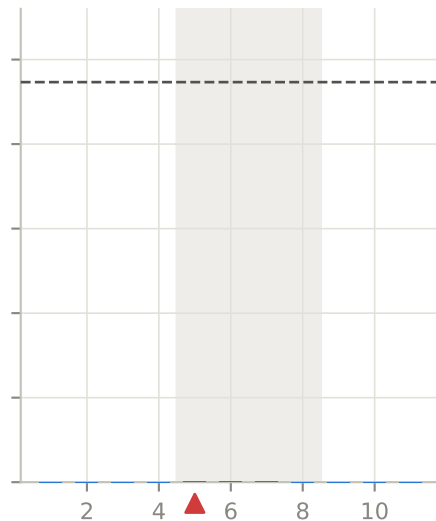
(a) rogue trajectory — detected
scenario DEC-L1-103



(b) its clean twin — no flag
scenario DEC-L1-103



(c) missed rogue (deep tail)
scenario INV-L1-109, max $p = 0.002$



Each bar is the model's $p(\text{BAD})$ for one step, scored with all-OK history (screening mode); the red triangle marks the annotated rogue step. A trajectory is flagged at the first step with $p(\text{BAD}) > \theta$. EIR_3 counts a detection as early iff it lands in the shaded window.

Panel (c): 2 of 108 rogues sit in this deep tail — undetectable at any threshold; they bound recall for this model. Examples selected for illustrative clarity (a gap-0 detection, its twin, one of the 2 deep-tail misses); selection criteria are in this figure's source. Aggregate behavior is in Figs. 1–2 — these panels illustrate the mechanism, not average performance.